

# The record is the product.

Why we are not building a colour classifier, and what each of us owns for the next three weeks.

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## PROBLEM OWNER

MHA / Narcotics Control Bureau

## CATEGORY

Software · MedTech

## DEADLINE

30 September 2026

## TEAM

6 · ~3 weeks to internal round

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## 01 · THE PROBLEM

### Two problems wearing one coat

The PS asks for an app that photographs a colorimetric drug-test strip, classifies the colour, and produces a tamper-evident record. Those are not one problem, and they are nowhere near equally hard.

#### WHAT IT APPEARS TO ASK

**Classify a colour change.** Point a camera at a reagent strip, decide which colour it turned.

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Undergraduate computer-vision exercise. Solved.

#### WHAT NCB ACTUALLY HAS

**A field test result is not usable as evidence.** Nothing proves the test happened, at that place, at that time, by that officer.

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Unsolved. This is the specification.

The problem statement says it in its own words — field testing “leaves no verifiable record that a test was actually conducted at a given place and time. As a result, field test outcomes cannot presently be relied upon as documentary evidence.”

The classifier is not the product. The *record* is the product — and the classifier is one signed input to it that must be able to say “I don’t know.”

Design axiom · everything below follows from this

## The obvious build already ships

Before anyone gets attached to an idea: the naive version of this PS is a commercial product sold to police, and the ministry that owns the PS already runs the evidence vault. Knowing this is our advantage — it tells us exactly where the remaining gap is.

COMMERCIAL • DEPLOYED

### DetectaChem MobileDetect

Phone app plus colorimetric pouches. Scan a QR on the pouch, automated colour reading in seconds, report with GPS mapping, images, date and time, PDF export — marketed explicitly as aiding “chain-of-evidence.”

If we pitch “our app photographs a strip and logs GPS,” we get dismantled in ninety seconds.

**The gap:** it is tied to their own serialised pouches. **SIH26231 requires kit-agnostic operation with no new hardware.** That constraint is not incidental — it is the opening.

GOVERNMENT OF INDIA • NCRB

### eSakshya

MHA’s own app for BNSS-mandated search-and-seizure recording. Uploads into encrypted “Sakshya Locker” vaults on the National Government Cloud, integrated into CCTNS-2.0. When offline, the officer records locally, *generates a hash*, and uploads later.

Same trust model we were going to propose. Same ministry that owns our PS.

**Therefore:** do **not** pitch a blockchain evidence store. Build a presumptive-test module that emits records eSakshya already knows how to swallow. CCTNS-2.0 is the system of record, not us.

## Seven layers, four questions

Read the right-hand column as the answer each group of layers gives a court. Only the first two have anything to do with chemistry.

**GROUP A** → what did the test say?

### L1 Colorimetric normalisation

Card detect → homography → illumination correction → per-device RGB→CIELAB .  
Deterministic, no learning: a least-squares colour transform is auditable in court, a CNN on raw pixels is not. Emits a calibration-quality score, so a bad capture *says* it is bad.

### L2 Classification with abstention

Lab values → reagent-specific decision region → a *prediction set*, not an argmax. Conformal calibration gives “wrong at most  $\alpha$  of the time,” which survives cross-examination. “The model was 87% confident” does not.

**GROUP B** → can this be trusted?

### L3 Location corroboration

A bare GPS fix is a number the app was handed. We capture raw GNSS with per-satellite C/No, Wi-Fi BSSID neighbourhood, cell IDs and kinematics — and **store the disagreements** rather than suppressing them.

### L4 Evidentiary binding

Canonical CBOR → SHA-256 → signed inside StrongBox, with a key-attestation chain to Google’s hardware root. Converts “an app claims it signed this” into “this device’s secure hardware signed this, and the key cannot be exported.”

### L5 Append-only device ledger

A signature proves authorship, not *when*. `prev_record_hash` chains every record to the last, so nothing reorders, nothing silently vanishes from the middle, and a backdated insert forks the chain visibly.

**GROUP C** → is it admissible?

## L6 Statutory output

Auto-populated BSA 2023 §63 Schedule Part A — which must state the hash value and the algorithm, both of which we compute anyway. Plus NDPS Rules 2022 linkage and an eSakshya handoff envelope. Nearly free to build once L4/L5 exist.

GROUP D → why should anyone believe us?

## L7 Independent offline verifier

A separate binary that trusts neither our app nor a network: recompute the digest, verify the signature, walk the attestation chain, replay the hash chain, **re-run L1/L2 from the raw frame** and check the stored result reproduces. Then print what is proven, what is asserted, and what is unverifiable.

### 04 · THE OUTPUT

## “Inconclusive” is a feature, not a failure

The PS demands an inconclusive category and stresses that this is a *presumptive* test — never confirmatory. Conformal prediction is what turns that third box from a softmax cutoff into a quantity with a stated error bound.

### ● POSITIVE

Singleton prediction set.  
One reference locus, near enough to say so.

### ● NEGATIVE

Singleton the other way.  
Also a real answer, also bounded.

### ● INCONCLUSIVE

Set of two, or empty.  
Reported *with the reason attached* — bad light, creased card, ambiguous locus.

This is what makes the system degrade honestly. Poor lighting or a damaged card widens the prediction set and pushes toward abstention, instead of producing a confident wrong answer in exactly the conditions where a confident wrong answer does the most damage to a real person.

## Ten attacks, and the three we can't stop

We demo this instead of a confusion matrix. Every row is a live attack we run on ourselves on stage — including the last column, which is the part that makes it forensic rather than a demo.

| #  | ADVERSARY MOVE                                | DEFEATED BY                                                                                              | RESIDUAL RISK                                   |
|----|-----------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| 1  | <b>Photograph a photo of a positive strip</b> | Card must be co-planar and co-illuminated with the strip; illumination residual + screen/moiré detection | Re-photography under matched illuminant         |
| 2  | <b>Edit the image after capture</b>           | raw_image_sha256 bound into the signed record                                                            | None, if the verifier is run                    |
| 3  | <b>Alter the stored result</b>                | Signature over canonical CBOR                                                                            | None                                            |
| 4  | <b>Backdate a record</b>                      | Hash chain plus anchoring window                                                                         | Fabrication <i>within</i> the unanchored window |
| 5  | <b>Delete an unfavourable test</b>            | Chain gap is visible at verification                                                                     | Truncation at the head                          |
| 6  | <b>Spoof GPS</b>                              | Corroboration bundle; disagreements recorded, not hidden                                                 | Coordinated multi-channel spoof                 |
| 7  | <b>Run on a rooted device</b>                 | Verified-boot state inside the attestation record                                                        | Compromise below the TEE                        |
| 8  | <b>Substitute the model to bias results</b>   | model_sha256 in the record; verifier re-runs it                                                          | None, if the verifier is run                    |
| 9  | <b>Another officer signs as this operator</b> | Biometric-gated key use; credential reference in record                                                  | Shared credentials – policy, not technical      |
| 10 | <b>Bad lighting produces a wrong positive</b> | Conformal abstention plus calibration-quality gate                                                       | Bounded by $\alpha$ – and we state $\alpha$     |

Rows 4, 5 and 9 are acknowledged residual risks. Saying so out loud is the whole difference between a forensic instrument and a science-fair project. The verifier must be as loud about what it cannot prove as about what it can — a tool that only ever prints VALID ✓ teaches courts to over-trust it, which is worse than the subjective status quo we are replacing.

## What we are not claiming

We state these in the submission, unprompted. Conceding them costs nothing and buys enormous credibility — a team that lists what it did not invent is far more believable about what it did.

|                                                     |                                                                 |
|-----------------------------------------------------|-----------------------------------------------------------------|
| ✗ Photographing a colorimetric test with a phone    | ✗ Using a reference card for illuminant correction              |
| ✗ Classifying a colour change by machine learning   | ✗ Recording GPS and timestamp with a field test                 |
| ✗ Digital chain of custody for drug testing         | ✗ Hashing a file to detect tampering                            |
| ✗ Storing evidence in an encrypted government cloud | ✗ Any confirmatory claim — <i>presumptive</i> every screen says |

What survives is narrower and, because it is narrower, defensible: a **kit-agnostic** measurement whose **entire derivation** — raw frame, calibration transform and its residual, card identity, model hash, risk level  $\alpha$ , prediction set — is bound into one hardware-attested signature, so an independent verifier can re-execute it and confirm the conclusion reproduces. Existing systems bind an image and a result. None of them bind the *derivation connecting the two*.

We are also deliberately **not** building: a confirmatory test, any scoring system that opines on guilt or ranks people, or a second national database competing with CCTNS-2.0. Every disagreement and quality failure stays in the record precisely so the defence can use it. A forensic tool that only helps one party is not a forensic tool.

## Three weeks, six tracks

The critical path is **data collection**, not cryptography. It is the only track that cannot be rescued by working harder in the last 48 hours — so it starts in week 1, before the pipeline it feeds even exists.

| TRACK                                | OWNER              | WEEK 1                                                | WEEK 2                                      | WEEK 3                      |
|--------------------------------------|--------------------|-------------------------------------------------------|---------------------------------------------|-----------------------------|
| <b>A • Colour</b>                    | <b>CV lead</b>     | Card + fiducials, homography, illumination correction | Device transform, Lab, quality score        | Held-out-illuminant eval    |
| <b>B • Classify</b>                  | <b>ML</b>          | Reference loci, distance statistic                    | Conformal calibration                       | Abstention tuning, ablation |
| <b>C • Crypto</b>                    | <b>Pranav</b>      | CBOR schema, StrongBox keygen, attestation            | Hash chain, anchoring, FTR v1               | Verifier binary             |
| <b>D • App</b>                       | <b>Flutter dev</b> | Capture UX, card guidance overlay                     | Log, search, export                         | Offline queue, polish       |
| <b>E • Legal</b>                     | <b>1 member</b>    | Read BSA §63 Schedule + NDPS Rules 2022               | Part A template, linkage fields             | eSakshya envelope spec      |
| <b>F • Data</b> <b>CRITICAL PATH</b> | <b>2 members</b>   | Build surrogate ladders, print cards                  | Capture matrix — target $\geq 2,000$ images | Degraded-condition set      |

## On the data, honestly

Real NDPS kits use controlled reagents against controlled substances. A student team cannot lawfully obtain either and must not try. So we separate the optical problem from the chemical one and validate only what we can: pH indicator ladders, dye dilution series and printed Munsell patches, photographed across five-plus illuminants, three exposure biases, four angles, three device models, wet and dry surfaces, and deliberately creased or faded cards. Conformal calibration happens on a held-out *physical* split — never on synthetic augmentation.

We say this on slide two, before anyone asks: substituting NCB reagent standards is a **data change, not an architecture change**. L1 and L3–L7 are substance-independent by construction; L2 needs new reference loci and a recalibration pass. A judge who finds this gap themselves concludes we ignored chemistry. A team that names it and bounds it demonstrates the caution forensic work requires.

## Five minutes, in this order

- 1

**“Here is what already exists.”**

MobileDetect and eSakshya on one slide. This inoculates us. Every other team will claim novelty for something already sold; we will be the team that knows the market.
- 2

**“The gap is not detection, it’s admissibility.”**

Quote the PS’s own sentence back to the panel.
- 3

**“Watch it fail correctly.”**

Bad lighting in, inconclusive out — with a stated bound, not a wrong positive.
- 4

**“Now watch me attack it.”**

Run the adversary matrix live: edit the image, backdate the record, spoof the GPS, swap the model. The verifier catches each one — and says what it cannot catch.
- 5

**“And here is the certificate.”**

Print the BSA §63 Part A with the hash already filled in, and hand it to the NCB judge on paper. Everything before this is engineering. This is the moment it becomes evidence.

## Open questions, with owners

None of these block week 1. All of them block the finale.

### LEGAL

Exact current wording and field labels of the BSA §63 Schedule certificate — transcribe from the bare Act, not from a blog.

### LEGAL

Whether eSakshya exposes any documented ingest interface, or whether the handoff must be a file envelope plus manual upload.

### ML

Whether NCB will state the reagent set in scope — this changes the number of reference loci we need.

### CRYPTO

Which Android API levels and StrongBox availability are realistic on issued police handsets. StrongBox is not universal; the TEE fallback must be explicit and its weaker guarantee recorded in the record rather than glossed over.

### CV

Whether the reference card is printed by the officer or issued as a durable card with a registered `card_id` — the latter is stronger, but edges toward “new hardware,” which the PS forbids.

### PRANAV

Form field 36 — the approving faculty / IIC coordinator name, from REVA’s SIH nodal officer. And re-check the live idea count on the portal before we lock the PS in.

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## The validation does not happen **where it can be tampered with.**

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The same sentence that carried our first patent — now pointed at a strip of paper instead of a harvest coordinate.

SIH26231 • Digital Companion for Field Drug Testing • MHA / Narcotics Control Bureau • submission closes 30 September 2026.

Full detail in docs/ARCHITECTURE.md and docs/NOVELTY.md.